Bourgain-root counterexamples to sequential Buo completeness of ℓ_p

Run `fa_banach_001`, lane 7

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Status

This packet records a candidate counterexample to the ℓ_p bullet in Question 5.2 of Kania and Swaczyna's paper *Bourgain–uo sequential completeness in vector lattices* [2]. The source asks whether Buo-Cauchy sequences converge in ℓ_p for $1 < p < \infty$. We show that the answer is negative for every $1 < p < \infty$.

The construction uses Bourgain's 1981 counterexample in ℓ_1 [1] and a coordinatewise p-root transform. The new point is that this transform preserves the relevant subsequence-difference Cauchy condition while turning ℓ_1-domination into ℓ_p-domination.

Source Question

The open question appears on page 12 of the source PDF.

5. Open problems

Question 5.1. Characterise those Banach lattices for which every norm bounded Buo-Cauchy sequence is uo-convergent. How does this property relate to the bounded uo-completeness studied in [9]?

Question 5.2. Do Buo-Cauchy sequences converge in

- ℓ_p for $1 < p < \infty$?
- the Morrey spaces $\mathcal{M}^{p,\kappa}(\mathbb{T})$ and in Besov/Triebel–Lizorkin spaces $B^s_{p,q}(\mathbb{T}), F^s_{p,q}(\mathbb{T})$ in the usual parameter ranges?

The relevant bullet asks whether Buo-Cauchy sequences converge in ℓ_p for $1 < p < \infty$.

Bourgain's Input

Bourgain considers on ℓ_1 the convergence given by coordinatewise convergence plus domination by a single element of ℓ_1. For sequences in ℓ_1, this is exactly order convergence. He calls a sequence Cauchy when the consecutive differences along every increasing subsequence converge in this sense, and proves that there is such a Cauchy sequence which is not convergent in this sense [1, p. 175].

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NON-COMPLETENESS OF SOME CONVERGENCE ON l^1

BY

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The aim of this paper is to solve the following problem posed by J. Mikusiński [1]:

In the space l^1 of absolutely summable sequences of real numbers we consider p-convergence: $x^n \twoheadrightarrow x$. This means that

(a) $\lim\limits_n x_i^n = x_i$ for every coordinate i,

(b) in l^1 there is some y such that $|x_i^n| \leqslant y_i$.

A sequence (x_n) is called *p-Cauchy* iff $x^{n_{k+1}} - x^{n_k} \twoheadrightarrow 0$ whenever (n_k) is an increasing sequence of integers. Is every p-Cauchy sequence p-convergent to some vector in l^1?

The answer is negative. We are going to show it by constructing a p-Cauchy sequence which is not p-convergent.

Denote by $\|\cdot\|_1$ the usual norm on l^1, thus

$$\|x\|_1 = \sum_{i=1}^{\infty} |x_i| \quad \text{for } x = (x_i).$$

We first give a finite reformulation of the problem. It is convenient to introduce the following notation: If $x^1, x^2, \ldots, x^N$ is a finite set of vectors in l^1, define

$$\alpha(x^1, x^2, \ldots, x^N) = \sum_i \max_{1 \leqslant n \leqslant N} |x_i^n|$$

and

$$\beta(x^1, x^2, \ldots, x^N) = \max_{1 \leqslant n \leqslant N} \|x^n\|_1 + \max_{1 \leqslant n_1 < n_2 < \ldots \leqslant N} \alpha(x^{n_1} - x^{n_2}, x^{n_2} - x^{n_3}, \ldots).$$

Clearly, $\beta(x^1, x^2, \ldots, x^N) \leqslant 3\alpha(x^1, x^2, \ldots, x^N)$.

[1] P 1190 in Colloquium Mathematicum 43 (1980), p. 388.

In the terminology of Kania and Swaczyna, Bourgain's result gives a Buo-Cauchy sequence (z_n) in ℓ_1 which does not converge in order in ℓ_1.

Proof Intuition

The obstruction in Bourgain's example is not tied to the linear structure of ℓ_1; it is tied to domination. The coordinatewise map

$$\Phi_p(t) = \operatorname{sgn}(t)|t|^{1/p}$$

converts ℓ_1-summability of a positive sequence into ℓ_p-summability of its p-root. It is also Hölder: small dominated differences in ℓ_1 become small dominated differences in ℓ_p. Thus Bourgain's Cauchy property survives the transform. Conversely, if the transformed sequence were order convergent in ℓ_p, raising coordinates back to the pth power would make Bourgain's original sequence order convergent in ℓ_1, which is impossible.

A Root-Map Lemma

Lemma 1. *Let $1 < p < \infty$, and define*

$$\Phi_p(t) = \operatorname{sgn}(t)|t|^{1/p} \qquad (t \in \mathbb{R}).$$

If $a_m - b_m \to 0$ in order in ℓ_1, then

$$\Phi_p(a_m) - \Phi_p(b_m) \to 0$$

in order in ℓ_p, where Φ_p is applied coordinatewise.

Proof. For real s, t one has

$$|\Phi_p(s) - \Phi_p(t)|^p \le 2^{p-1}|s - t|.$$

Indeed, if s and t have the same sign this follows from the $1/p$-Hölder continuity of $u \mapsto u^{1/p}$ on $[0, \infty)$. If they have opposite signs, then concavity gives

$$|\Phi_p(s) - \Phi_p(t)| = |s|^{1/p} + |t|^{1/p} \le 2^{1-1/p}(|s| + |t|)^{1/p} = 2^{1-1/p}|s - t|^{1/p}.$$

Since $a_m - b_m \to 0$ in order in ℓ_1, there are positive ℓ_1-vectors $w_\alpha \downarrow 0$ such that, for every α,

$$|a_m - b_m| \le w_\alpha$$

eventually in m. The displayed scalar estimate gives

$$|\Phi_p(a_m) - \Phi_p(b_m)| \le 2^{1-1/p} w_\alpha^{1/p}$$

eventually, where the root is taken coordinatewise. Since $(w_\alpha^{1/p})^p = w_\alpha \in \ell_1$, we have $w_\alpha^{1/p} \in \ell_p$, and $w_\alpha^{1/p} \downarrow 0$ coordinatewise. Thus the transformed differences converge to 0 in order in ℓ_p. $\square$

Counterexample

Theorem 1. *For every $1 < p < \infty$, ℓ_p is not sequentially* Buo*-complete. Equivalently, there is a* Buo*-Cauchy sequence in ℓ_p which does not converge in order, and hence does not* Buo*-converge.*

Proof. Let $(z_n) \subset \ell_1$ be Bourgain's sequence: it is Cauchy for the subsequence-difference convergence described above, but it is not convergent for coordinatewise-plus-ℓ_1-dominated convergence. Since that convergence is order convergence in ℓ_1, (z_n) is Buo-Cauchy in ℓ_1 but not order convergent.

Fix $1 < p < \infty$, and define

$$x_n(j) = \Phi_p(z_n(j)) = \operatorname{sgn}(z_n(j))|z_n(j)|^{1/p} \qquad (j \in \mathbb{N}).$$

Then $x_n \in \ell_p$, because

$$\|x_n\|_p^p = \sum_j |x_n(j)|^p = \sum_j |z_n(j)| = \|z_n\|_1 < \infty.$$

We first show that (x_n) is Buo-Cauchy in ℓ_p. Let (n_k) be any strictly increasing sequence. Since (z_n) is Buo-Cauchy in ℓ_1,

$$z_{n_{k+1}} - z_{n_k} \to 0$$

in order in ℓ_1. Applying Lemma 1 with $a_k = z_{n_{k+1}}$ and $b_k = z_{n_k}$, we obtain

$$x_{n_{k+1}} - x_{n_k} = \Phi_p(z_{n_{k+1}}) - \Phi_p(z_{n_k}) \to 0$$

in order in ℓ_p. Since (n_k) was arbitrary, (x_n) is Buo-Cauchy in ℓ_p.

It remains to show that (x_n) is not order convergent in ℓ_p. Suppose otherwise. Then $x_n \to x$ in order for some $x \in \ell_p$. Order convergence in ℓ_p implies coordinatewise convergence and eventual domination: after discarding finitely many terms, there is $y \in (\ell_p)_+$ with $|x_n| \le y$. Define

$$\Psi_p(t) = \operatorname{sgn}(t)|t|^p.$$

Coordinatewise, $z_n = \Psi_p(x_n)$, and $z_n(j) \to \Psi_p(x(j))$. Moreover

$$|z_n(j)| = |x_n(j)|^p \le y(j)^p,$$

and $y^p \in \ell_1$, for all sufficiently large n. Adding the finitely many initial vectors $|z_1|, \dots, |z_{N-1}|$ to this dominator gives one element of ℓ_1 dominating the whole sequence. Therefore (z_n) converges in the coordinatewise-plus-ℓ_1-dominated sense, equivalently in order in ℓ_1. This contradicts Bourgain's choice of (z_n).

Thus (x_n) is a Buo-Cauchy sequence in ℓ_p which does not converge in order. □

Compatibility with the Norm-Convergence Packet

A separate packet in this run proves that every Buo-Cauchy sequence in ℓ_p, $1 \le p < \infty$, is norm Cauchy and hence norm convergent. The present counterexample is compatible with that result: it rules out the stronger conclusion of order convergence, not norm convergence.

Novelty and Literature Check

The supporting literature used here is Bourgain's 1981 paper [1], located through Crossref DOI 10.4064/cm-44-1-175-178 and downloaded from the IMPAN publication page. Page 175 contains the relevant definition of the Cauchy condition and the negative answer in ℓ_1.

Searches for exact phrases around "Bourgain–uo", "Buo-Cauchy", "ℓ_p", and the source question surfaced the source paper and adjacent bounded-uo completeness material, but no later explicit resolution of the ℓ_p bullet. The counterexample is best viewed as a short transfer argument from Bourgain's ℓ_1 construction; novelty confidence is therefore moderate, and human review should check whether this root-map transfer is already known or implicit in unpublished notes.

Human Review Recommendation

Check two points:

1. Bourgain's Cauchy condition on page 175 is exactly the subsequence-difference order convergence condition used here.

2. The scalar Hölder estimate in Lemma 1 correctly transports order-null difference sequences from ℓ_1 to ℓ_p.

If both checks pass, the ℓ_p bullet of Question 5.2 is answered negatively for every $1 < p < \infty$.

References

[1] J. Bourgain, *Non-completeness of some convergence on ℓ^1*, Colloquium Mathematicum **44** (1981), no. 1, 175–178, DOI: 10.4064/cm-44-1-175-178.

[2] T. Kania and J. Swaczyna, *Bourgain–uo sequential completeness in vector lattices*, arXiv:2512.14949, 2025.